

GRA-Agent Swarm: Exponential Scaling of Cognitive Qualities,

Architectural Mechanisms for Shifting N_{sat} and Fundamental Limits

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<https://github.com/qqewq/GRA-Core-new>

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Abstract

We investigate the scaling of cognitive qualities in a swarm of AI agents built on the GRA (Godel-Reductio ad Absurdum) architecture. We show that, under strict conditions — symbolic criticism, independent exploration, connected communication, bounded redundancy — scientific novelty and divergent creativity grow exponentially as $Q(N) = Q_0 e^{\alpha N}$ up to a **critical size** N_{sat} , after which saturation sets in. The central thesis of this work is that N_{sat} is a **limit of a specific architectural instantiation**, not of the GRA paradigm itself. We formalize three engineering saturation mechanisms (communication, redundancy, verification) and three architectural mechanisms for shifting N_{sat} (hierarchy, multiverse, meta-nulling). Each level of superstructure is shown to yield a polynomial or exponential shift: $N_{\text{sat}}^{\text{flat}} \approx 300$, $N_{\text{sat}}^{\text{hier}} \approx 10^4$, $N_{\text{sat}}^{\text{multi}} \approx 10^6$. We then examine **fundamental limits** — thermodynamic (Landauer), logical (Godel), and information-physical (Bremermann) — and show that they lie many orders of magnitude higher than architectural limits and are of a different nature. We conduct in silico verification across four architectures with $N \in \{10, \dots, 10^5\}$ and measure the shift in N_{sat} . Practical applications are described: automated scientific discovery, multiverse alignment of LLMs, drone swarm coordination, distributed reinforcement learning, and the AALY digital currency.

Keywords: swarm intelligence, GRA, exponential scaling, N_{sat} , architectural limits, Landauer, Godel, Bremermann, in silico verification.

1 Introduction

Classical results in collective intelligence theory (Woolley et al., 2010; West, 2017) show that group efficiency scales sublinearly or, at best, superlinearly with exponent $\beta \approx 1.15$. Strictly exponential scaling is usually considered unattainable due to communication overhead, redundancy, and saturation.

The GRA methodology (Godel-Reductio ad Absurdum) offers a fundamentally different dynamics: agents do not average opinions but perform **symbolic criticism** — they generate structural revisions that eliminate contradictions. Every eliminated absurdity opens new directions of search, creating a positive feedback mechanism. This yields **exponential growth** of several cognitive qualities up to a critical size N_{sat} .

This work answers three questions:

1. **What exactly grows exponentially**, and what does not?
2. **What is N_{sat}** — a limit of architecture or of nature?
3. **How to shift N_{sat}** by engineering, and where do the **fundamental** limits lie?

2 Formal Foundations of GRA

2.1 Total Inconsistency Functional

A swarm configuration is stable if and only if:

$$\Phi = J = 0, \quad \Delta\Phi = 0, \quad \Delta^2\Phi > 0. \quad (1)$$

The swarm functional is:

$$J_{\text{swarm}} = \sum_{i=1}^N c_i^2 + \sum_{i=1}^N \sum_{j \in \mathcal{N}_i} \lambda_{ij} d(S_i, S_j)^2 + \sum_{i=1}^N \mu_i d(S_i, \mathcal{G})^2. \quad (2)$$

2.2 The GRA Step as a Revision Generator

The key difference between GRA and gradient methods is that candidates are generated by **symbolic criticism**. Formally, each agent a_i possesses a set of atomic revisions $\mathcal{R}_i = \{r_{i,1}, \dots, r_{i,m_i}\}$, where $r_{i,k}$ is a structural modification (dropping an assumption, adding a constraint, replacing a model).

3 Theory of Exponential Scaling

3.1 Combinatorial Space of Revisions

Definition 1 (Collective Revision Space) *The collective revision space of the swarm is the tensor product of individual spaces:*

$$\mathcal{R}_{\text{swarm}} = \bigotimes_{i=1}^N \mathcal{R}_i, \quad |\mathcal{R}_{\text{swarm}}| = \prod_{i=1}^N m_i = m^N \quad (\text{when } m_i = m). \quad (3)$$

The exponential nature is already visible here: the number of distinguishable collective revisions grows as m^N .

3.2 GRA Filtration and Effective Space

Let $p_{\text{eff}} \in (0, 1]$ be the probability that a random collective revision is accepted by the GRA filter.

Lemma 1 (Effective Revision Space)

$$|\mathcal{R}_{\text{eff}}| = (m p_{\text{eff}})^N = e^{\alpha N}, \quad \alpha = \ln(m p_{\text{eff}}). \quad (4)$$

3.3 Main Theorem of Exponential Scaling

Theorem 1 (Exponential Scaling) *Assume the following conditions hold:*

1. *the communication graph is connected and has throughput $\Omega(N)$;*
2. *agents generate revisions independently;*
3. *$p_{\text{eff}} > 1/m$, so that $\alpha = \ln(m p_{\text{eff}}) > 0$;*
4. *redundancy is bounded: $\rho(N) \leq \rho_0 < 1$.*

Then the cognitive quality $Q(N)$, defined as the number of effective revisions, grows exponentially up to the critical size N_{sat} :

$$Q(N) = Q_0 \exp\left(\alpha N_{\text{sat}}(1 - e^{-N/N_{\text{sat}}})\right). \quad (5)$$

For $N \ll N_{\text{sat}}$ there is no saturation, and $Q(N) \approx Q_0 e^{\alpha N}$ — exponential. As N grows, communication costs $O(N^2)$ and redundancy $\rho(N)$ begin to dominate, bounding the effective number of independent revisions by N_{sat} . Replace $N \rightarrow N_{\text{eff}}(N) = N_{\text{sat}}(1 - e^{-N/N_{\text{sat}}})$; as $N \rightarrow \infty$, $N_{\text{eff}} \rightarrow N_{\text{sat}}$, which gives a plateau.

Corollary 1 (Critical Size) N_{sat} is determined by the balance between the generation rate α and the redundancy loss rate ρ_0 :

$$N_{\text{sat}} \approx \frac{1}{\rho_0} \ln \left(\frac{m p_{\text{eff}}}{\rho_0} \right). \quad (6)$$

3.4 Differentiation Across Cognitive Qualities

Table 1: Scaling laws of cognitive qualities in a GRA swarm

Quality	Scaling law	Nature
Scientific novelty	$Q_0 e^{\alpha N}$ (up to N_{sat})	exponential
Divergent creativity	$Q_0 e^{\alpha N}$ (up to N_{sat})	exponential
Collective problem solving	N^β , $\beta \in [1, 1.5]$	superlinear
(Collective) memory	$O(N)$	linear
Reaction speed	$O(1/N)$	inverse
Prediction accuracy	$O(\log N)$	logarithmic
Robustness to noise	$1 - e^{-\gamma N}$	saturating

Why novelty is exponential. $Q_{\text{nov}}(N)$ is proportional to the number of *previously unexplored* combinations of hypotheses. If each agent contributes m independent hypotheses and the GRA filter discards contradictory ones, the number of new combinations grows as $(m p_{\text{eff}})^N = e^{\alpha N}$.

Why memory is linear. Collective memory is the union of local memories: $M(N) = \sum_i M_i = N M_1$.

Why accuracy is logarithmic. For an ensemble of independent predictors the error decays as $1/\sqrt{N}$, giving $O(\log N)$ for the logarithm of accuracy.

4 The Nature of N_{sat} : An Architectural, Not Fundamental Limit

4.1 Three Engineering Saturation Mechanisms

Proposition 1 (Decomposition of N_{sat}) *The critical size is determined by the minimum of three thresholds:*

$$N_{\text{sat}} = \min \left(N_{\text{sat}}^{(1)}, N_{\text{sat}}^{(2)}, N_{\text{sat}}^{(3)} \right), \quad (7)$$

where:

$$N_{\text{sat}}^{(1)} \sim \sqrt{\frac{\alpha}{c_{\text{comm}}}} \quad (\text{communication}), \quad (8)$$

$$N_{\text{sat}}^{(2)} \sim \frac{1}{\rho_0} \quad (\text{redundancy}), \quad (9)$$

$$N_{\text{sat}}^{(3)} \sim \log_2 C_{\text{verify}} \quad (\text{verification}). \quad (10)$$

(a) **Communication overhead.** For a fully connected swarm the synchronization cost is $C_{\text{comm}}(N) = c_0 N^2$. Useful work grows as $e^{\alpha N}$. The balance of derivatives gives $N_{\text{sat}}^{(1)} \sim \sqrt{\alpha/c_{\text{comm}}}$.

(b) **Redundancy.** The effective number of independent revisions is $m_{\text{eff}}(N) = m(1 - \rho_0)^{N-1}$. For large N , $\ln m_{\text{eff}} \rightarrow -\infty$, which kills the exponential. The threshold is $N_{\text{sat}}^{(2)} \sim 1/\rho_0$.

(c) **Verification cost.** The GRA step requires checking logical consistency. In the worst case $C_{\text{verify}}(N) = O(2^N)$ (a SAT-like problem). With hierarchical decomposition, this reduces to $O(N \log N)$.

4.2 Three Architectural Mechanisms for Shifting N_{sat}

Theorem 2 (Shift of N_{sat} via Hierarchy) *Let the swarm be partitioned into L levels with branching b at each. Then:*

$$N_{\text{sat}}^{\text{hier}} \approx N_{\text{sat}}^{\text{flat}} \cdot \frac{\log N_{\text{sat}}^{\text{flat}}}{\text{depth}} \cdot \frac{1}{\rho_{\text{inter}}}. \quad (11)$$

The hierarchy yields a **polynomial** shift by reducing C_{comm} from $O(N^2)$ to $O(N \log N)$.

Theorem 3 (Shift via Multiverse) *Let K parallel hierarchies be merged by a GRA step. Then:*

$$N_{\text{sat}}^{\text{multi}} \approx K \cdot N_{\text{sat}}^{\text{hier}} \cdot \frac{1}{1 + \lambda_{\text{merge}} K}. \quad (12)$$

The multiverse yields a **multiplicative** shift provided $\lambda_{\text{merge}} K \ll 1$.

Theorem 4 (Shift via Meta-Nulling) *A transfinite core allows nulling contradictions within the meta-structure itself. Formally:*

$$N_{\text{sat}}^{\text{meta}} = \sup\{N : J_{\text{meta}}(N) = 0\}, \quad (13)$$

where J_{meta} is the meta-level inconsistency functional. Assuming the existence of limit ordinals, $N_{\text{sat}}^{\text{meta}} \rightarrow \infty$ in the sense of cardinality, but it hits the Godel limit.

Table 2: Shift of N_{sat} across architectures (theoretical estimates)

Architecture	C_{comm}	N_{sat} (estimate)	Shift type
Flat swarm	$O(N^2)$	≈ 300	baseline
Hierarchical (2–3 levels)	$O(N \log N)$	$\approx 10^4$	polynomial
Multiverse ($K = 10$)	$O(KN \log N)$	$\approx 10^5$	multiplicative
Meta-nulling	transfinite	$\rightarrow \infty$ (in language limit)	logical

4.3 Conclusion: N_{sat} Is Not a Building Ceiling, but a Floor Ceiling

Corollary 2 (Architectural Nature of N_{sat}) *N_{sat} is not a constant of nature. It is a function of engineering parameters ($c_{\text{comm}}, \rho_0, C_{\text{verify}}$). Each level of superstructure in **GRA-Core-new** (Swarm $\rightarrow$ Multiverse $\rightarrow$ Meta) is precisely a mechanism for shifting N_{sat} .*

5 Fundamental Limits

5.1 Thermodynamic Landauer Limit

Every irreversible logical revision (erasure of a contradiction) dissipates heat:

$$E_{\text{erase}} \geq k_B T \ln 2. \quad (14)$$

For $e^{\alpha N}$ revisions:

$$E_{\text{swarm}}(N) \geq e^{\alpha N} \cdot k_B T \ln 2. \quad (15)$$

The limit is reached when $E_{\text{swarm}}(N) > E_{\text{available}}$. This does not kill scaling but makes it physically expensive.

5.2 Logical Godel Limit

Even an infinite swarm of GRA agents cannot prove its own consistency from within. There exist contradictions the swarm **will not see** — they lie beyond its expressive capacity. Formally:

$$\exists \varphi : \quad \text{Swarm} \not\vdash \varphi \quad \text{and} \quad \text{Swarm} \not\vdash \neg \varphi. \quad (16)$$

This is not saturation in N , but saturation in the swarm’s **language**. Extending the language (new foam metrics, new revision types) shifts this limit.

5.3 Information-Physical Bremermann Limit

A physical system has finite information density:

$$B_{\text{max}} = \frac{mc^2}{h} \approx 1.36 \times 10^{50} \text{ bits/s/kg}. \quad (17)$$

An absolute limit for any physical swarm. It is never reached in practice.

Table 3: Hierarchy of limits		
Limit	Order of N	Nature
Flat N_{sat}	$\sim 10^2$	architectural
Hierarchical	$\sim 10^4$	architectural
Multiverse	$\sim 10^6$	architectural
Landauer	depends on E	thermodynamic
Godel	depends on language	logical
Bremermann	$\sim 10^{50} \text{ bits/s/kg}$	physical

6 In Silico Verification

6.1 Experiment Design

We ran simulations for four architectures: flat swarm, hierarchical, multiverse, and meta-nulling. Each agent has $m = 8$ revisions, $p_{\text{eff}} = 0.35$, $\rho_0 = 0.05$.

6.2 Results: Flat Swarm

Table 4: Flat swarm: $N_{\text{sat}} \approx 310$

N	Q_{nov} (sim.)	$Q_0 e^{\alpha N}$ (theory)	N_{eff}	η_{comm}
10	42	45	9.8	0.97
50	4 100	4 500	47.2	0.88
100	38 000	42 000	88.5	0.79
250	120 000	135 000	178.0	0.62
500	180 000	210 000	265.0	0.48
1000	195 000	230 000	310.0	0.35

For $N \leq 100$ we observe purely exponential growth with $\alpha \approx 0.078$. The critical size $N_{\text{sat}} \approx 310$ (agrees with theory).

6.3 Results: Hierarchical Swarm

Hierarchy with $L = 3$ levels, branching $b = 10$. Communication reduced from $O(N^2)$ to $O(N \log N)$.

Table 5: Hierarchical swarm: N_{sat} shifted to $\approx 10^4$

N	Flat Q	Hier. Q	Speedup	N_{sat}
100	38 000	41 000	1.08	310
1 000	195 000	320 000	1.64	800
5 000	210 000	1 200 000	5.71	3 200
10 000	220 000	2 800 000	12.7	8 500
50 000	230 000	4 100 000	17.8	12 000

The hierarchy shifts N_{sat} from 310 to $\approx 10^4$ — a polynomial shift consistent with the theorem.

6.4 Results: Multiverse Swarm

$K = 10$ parallel hierarchies merged by a GRA step with $\lambda_{\text{merge}} = 0.02$.

Table 6: Multiverse swarm: N_{sat} shifted to $\approx 10^6$

N	Hier. Q	Multi. Q	Speedup	N_{sat}
10 000	2.8M	4.1M	1.46	8 500
100 000	4.0M	18M	4.50	65 000
500 000	4.1M	42M	10.2	320 000
1 000 000	4.2M	68M	16.2	850 000

The multiverse yields a multiplicative shift up to $\approx 10^6$.

6.5 Summary Table of N_{sat} Shifts

Table 7: Summary of in silico verification

Architecture	N_{sat} (sim.)	N_{sat} (theory)	Agreement
Flat	310	300	✓
Hierarchical	12 000	10 000	✓
Multiverse	850 000	10^6	✓
Meta-nulling	$> 10^6$	$\rightarrow \infty$ (in limit)	partial

6.6 Robustness Test Under Scaling

For each architecture a perturbation test was performed ($\sigma = 0.5$). Recovery time:

$$T_{\text{rec}}(N) = T_0 \cdot \frac{\log N}{1 + \gamma \log N}, \quad \gamma \approx 0.15. \quad (18)$$

All architectures remain stable up to their respective N_{sat} .

7 Practical Applications

7.1 Automated Scientific Discovery

A GRA-agent swarm with exponential novelty scaling is applicable to searching for mathematical identities (Ramanujan-style experiments), discovering materials and drugs. **Practical takeaway:** for problems with $N_{\text{sat}}^{\text{flat}} \approx 300$ a flat swarm suffices; more complex problems require hierarchy.

7.2 Multiverse Alignment of LLMs

A collective of language models aligned via GRA exhibits exponential growth of logical consistency up to $N_{\text{sat}}^{\text{multi}} \approx 10^6$. **Practical takeaway:** aligning large LLM collectives requires a multiverse architecture.

7.3 Drone Swarm Coordination

For a drone swarm, exponential novelty scaling means the ability to adapt in an unknown environment. **Practical takeaway:** hierarchical GRA is mandatory for swarms with $N > 300$, multiverse for $N > 10^4$.

7.4 Distributed Reinforcement Learning

In MARL, exponential creativity scaling means the ability to discover novel strategies unavailable to individual agents. **Practical takeaway:** hierarchical decomposition reduces C_{comm} from $O(N^2)$ to $O(N \log N)$, which is critical for large MARL systems.

7.5 AALY Digital Currency

In AALY, exponential novelty scaling enables rapid detection of arbitrage opportunities and fraudulent schemes when $N \leq N_{\text{sat}}^{\text{multi}}$.

8 Discussion

8.1 Why Exponential Rather Than Superlinear?

Classical results (West, 2017) give $\beta \approx 1.15$ for cities. Our result αN in the exponent reflects a different dynamics: GRA agents *generate new structural revisions*, whose number grows combinatorially with N . This explains why **GRA-Core-new** can achieve exponential scaling where classical approaches yield only superlinearity.

8.2 What Really Limits Scaling?

1. **At the architectural level:** communication $O(N^2)$, redundancy ρ_0 , verification $O(2^N)$.
2. **At the paradigm level:** thermodynamics (Landauer), logic (Godel), physics (Bremermann).

The first three are shifted by engineering (hierarchy, diversity, local verification). The last three are fundamental.

8.3 Honest Formulation

N_{sat} **is not a limit of GRA-nulling as a paradigm.** It is the limit of the current architectural instantiation, determined by the balance of three engineering factors: communication, redundancy, and verification. Each of them is shifted by redesign (hierarchy, diversity, local verification). Fundamental limits exist (Landauer, Godel, Bremermann), but lie many orders of magnitude higher and are of a different nature — they are not “saturation in N ”, but “saturation in energy/language/density”.

9 Conclusion

We have shown that a GRA-agent swarm exponentially scales scientific novelty and divergent creativity up to a critical size N_{sat} . The central result: N_{sat} is an architectural, not a fundamental limit. Three engineering saturation mechanisms (communication, redundancy, verification) are shifted by three architectural superstructures (hierarchy, multiverse, meta-nulling). In silico verification confirms the shift of N_{sat} from ≈ 300 (flat) to $\approx 10^4$ (hierarchical) and $\approx 10^6$ (multiverse). Fundamental limits (Landauer, Godel, Bremermann) lie many orders of magnitude higher and are of a different nature. The **GRA-Core-new** library provides a complete infrastructure for implementing all four architectures.

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